

9.4.23

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Question

Find the roots of the quadratic equation graphically.

$$6x^2 - x - 2 = 0 \quad (1)$$

Solution

$$y = 6x^2 - x - 2 \quad (2)$$

This equation can be represented as the conic:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (3)$$

where

$$\mathbf{V} = \begin{pmatrix} 6 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}, \quad f = -2 \quad (4)$$

To find the roots, we find the points of intersection of the conic with the x-axis:

$$\mathbf{x} = \mathbf{h} + k_i \mathbf{m} \quad (5)$$

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (6)$$

Solution

Substitute into the conic equation:

$$(\mathbf{h} + k_i \mathbf{m})^\top \mathbf{V}(\mathbf{h} + k_i \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + k_i \mathbf{m}) + f = 0 \quad (7)$$

This simplifies to a quadratic in k_i :

$$k_i^2 \mathbf{m}^\top \mathbf{V} \mathbf{m} + 2k_i \mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) + g(\mathbf{h}) = 0 \quad (8)$$

Now substituting the values:

$$k_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - \mathbf{m}^\top \mathbf{V} \mathbf{m} \cdot g(\mathbf{h})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (9)$$

Solution

$$\Rightarrow k_i = \frac{-(-\frac{1}{2}) \pm \sqrt{(-\frac{1}{2})^2 - 6(-2)}}{6} = \frac{\frac{1}{2} \pm \sqrt{\frac{1}{4} + 12}}{6} = \frac{\frac{1}{2} \pm \frac{7}{2}}{6} \quad (10)$$

$$\Rightarrow k_1 = \frac{4}{6} = \frac{2}{3}, \quad k_2 = \frac{-3}{6} = -\frac{1}{2} \quad (11)$$

Hence, the real roots are:

$$\mathbf{x}_1 = \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix} \quad (12)$$

Code (C)

Part 1

```
#include <stdio.h>
#include <math.h>

void roots(double *root1, double *root2) {
    double a = 6.0, b = -1.0, c = -2.0;
    double discriminant = b*b - 4*a*c;

    if (discriminant >= 0) {
        *root1 = (-b + sqrt(discriminant)) / (2*a);
        *root2 = (-b - sqrt(discriminant)) / (2*a);
    } else {
        *root1 = NAN;
        *root2 = NAN;
    }
}
```

Part 1

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load the shared C library
lib = ctypes.CDLL("./code15.so") # or .dll on Windows

# Prepare C function
lib.roots.argtypes = [ctypes.POINTER(ctypes.c_double), ctypes.
    POINTER(ctypes.c_double)]
lib.roots.restype = None
```

Part 2

```
# Create C doubles to store the roots
root1 = ctypes.c_double()
root2 = ctypes.c_double()

# Call C function
lib.roots(ctypes.byref(root1), ctypes.byref(root2))

# Extract Python values
x1 = root1.value
x2 = root2.value

print("Roots from C:", x1, x2)
```


Part 3

```
# Plot the quadratic
x = np.linspace(min(x1, x2) - 1, max(x1, x2) + 1, 400)
y = 6*x**2 - x - 2

plt.figure(figsize=(8,6))
# Include the roots in the legend label
plt.plot(x, y, label=f'$y = 6x^2 - x - 2$\nRoots: x1={x1:.2f}, x2={x2:.2f}')
plt.axhline(0, color='black', linewidth=1)
plt.scatter([x1, x2], [0,0], color='red', zorder=5)
# Label the roots on the graph (optional)
offset_y = max(y)*0.05
offset_x = 0.02
```

Python Code: Plotting

Part 4

```
plt.text(x1 + offset_x, 0 + offset_y, f'x1({x1:.2f})', ha='left',  
         va='bottom', color='red')  
plt.text(x2 + offset_x, 0 + offset_y, f'x2({x2:.2f})', ha='left',  
         va='bottom', color='red')  
  
plt.xlabel('x')  
plt.ylabel('y')  
plt.title('Graphical solution of  $6x^2 - x - 2 = 0$ ')  
plt.legend()  
plt.grid(True)  
plt.tight_layout()  
  
# Save figure  
plt.savefig('fig15.png', dpi=300)  
plt.show()
```

Graphical Representation

